

WhisperMini: Optimizing Whisper ASR for Urdu

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

The primary goal of this research is to optimize OpenAI's Whisper ASR model for Urdu transcription, particularly for handling large datasets generated in high-demand environments such as call centers. Whisper, although highly effective for English and other widely spoken languages, struggles with low-resource languages like Urdu. This study aims to overcome these challenges by applying various optimization techniques to reduce computational demands without compromising the model's transcription accuracy.

In this report, we present the initial iteration of the optimization process, focusing on the proper evaluation of the Whisper model, its reverse engineering and an in-depth study of other relevant literature. This work sets the base for better understanding and optimization of the model which is an important step toward achieving the project's objectives.

1.1 Problem Domain

The domain of this study is Automatic Speech Recognition (ASR), an important part of various industries. Especially call centers which generate vast amounts of audio data and require transcription for analysis. Transcribing this data accurately and efficiently is essential for operations, customer support, and data analysis. However, existing ASR models, such as Whisper, are not well-suited for handling Urdu, a low-resource language. This research addresses the gap in optimizing ASR systems for real-time, large-scale transcription of Urdu speech data. Specifically, the major challenge is high computational costs of existing models and the language-specific nuances that reduce transcription accuracy.

1.2 Research Problem Statement

The research focuses on improving Whisper’s efficiency in transcribing Urdu, particularly for large-scale datasets. In our usecase we have approximately 6000 hours of call center audio data generated daily. Currently, Whisper transcribes at a 1:1 ratio, meaning that 6000 hours of data would take 6000 hours to transcribe, which is computationally expensive and impractical for real-time applications. Additionally, Whisper is not fine-tuned for Urdu’s linguistic and phonetic characteristics, further reducing its transcription accuracy. The core research problem is how to optimize Whisper to transcribe large datasets faster and with greater accuracy for Urdu, while minimizing resource usage.

1.3 Software Requirements Specification (if applicable)

This project involves the development of a specialized version of Whisper, termed Whisper-Mini, which incorporates several key optimizations:

- **Non-Functional Requirements:**

1. **Efficient Transcription:** The system should transcribe Urdu audio data more quickly than the baseline Whisper model.
2. **High Accuracy:** The transcription accuracy for Urdu should be at least maintained or increased through fine-tuning with Urdu-specific datasets.
3. **Web Interface:** The system should provide a user-friendly web interface for uploading and receiving transcribed audio.
1. **Performance:** The system should be able to transcribe large volumes of audio data with reduced computational resources.
2. **Scalability:** The system should handle transcription requests for thousands of hours of audio without degradation in performance.
3. **Accuracy:** The system should achieve a significant reduction in Word Error Rate (WER) and Character Error Rate (CER) for Urdu.

These initial specifications provide the groundwork for optimizing Whisper to meet the specific requirements of the target domain.

1.4 Current Analysis on Whisper

To evaluate Whisper’s performance on Urdu transcription, we conducted an initial analysis using a random sample of 50 audio clips from Mozilla Common Voice. Each audio file

was transcribed using Whisper’s large v3 model under two configurations: one with GPU acceleration (using CUDA) and another with default CPU processing. For this analysis, we utilized an AMD Ryzen 5 3600, a 6-core, 12-thread processor, alongside an NVIDIA RTX 3060 with 12 GB of VRAM. Given that the large Whisper model requires approximately 10 GB of VRAM, this setup provided an ideal scenario with ample resources, allowing us to observe Whisper’s efficiency under optimal conditions.

To measure performance, we tracked CPU, GPU, and RAM usage during transcription and calculated the average transcription time ratio (transcription time per audio duration) across configurations. Our exploratory data analysis (EDA), presented through visualizations, highlights Whisper’s computational demands and transcription efficiency, showcasing how transcription time scales with audio duration and the significant performance improvements afforded by GPU acceleration.

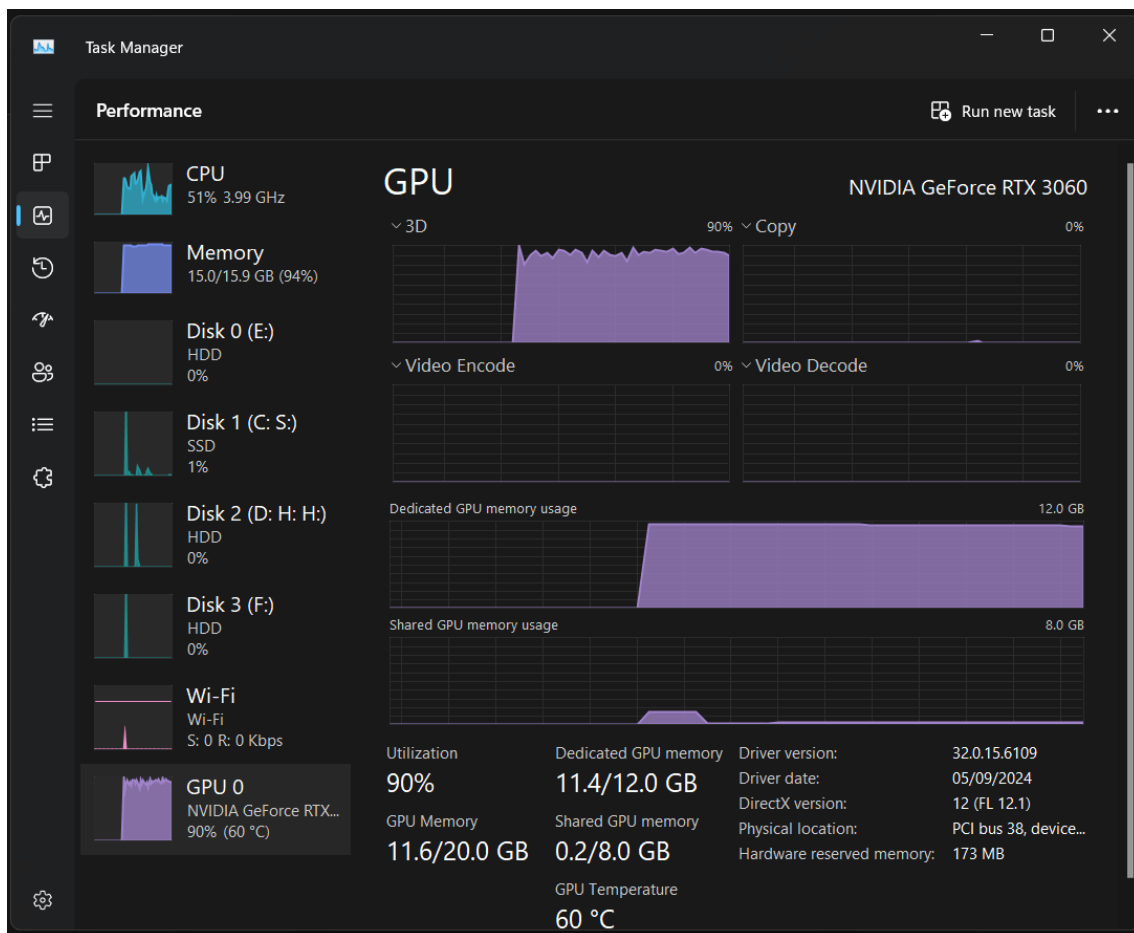


Figure 1.1: Resource usage while calculating transcription ratio with CUDA enabled

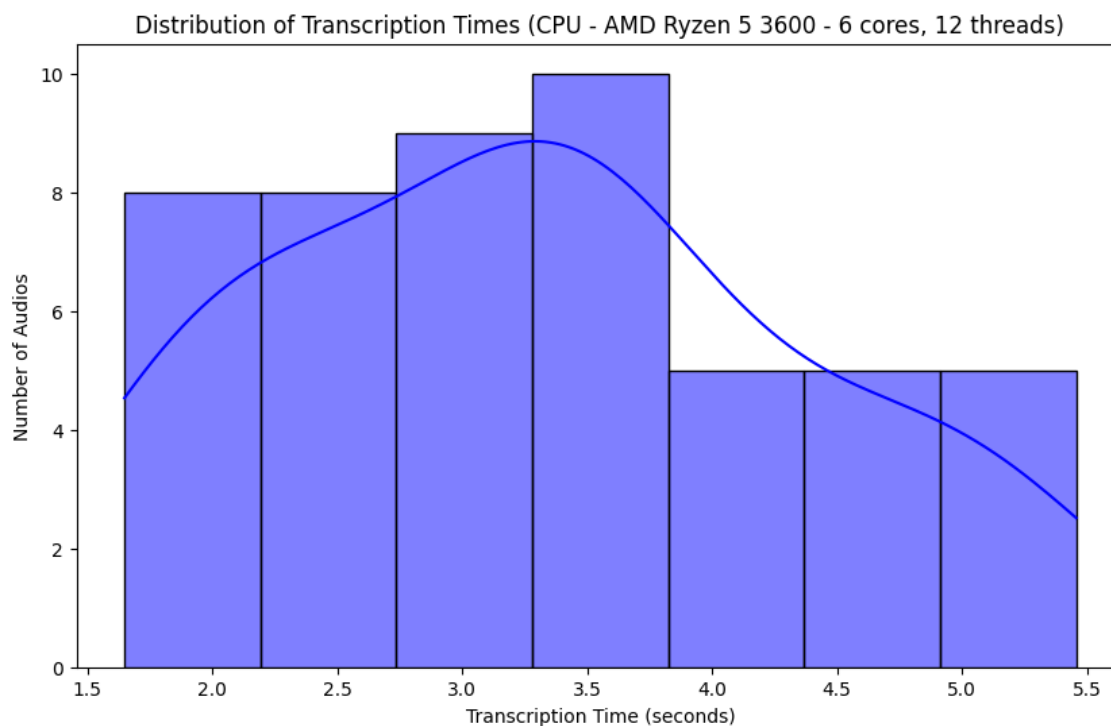


Figure 1.2: Transcription Times (CPU) compared to number of audios from sample

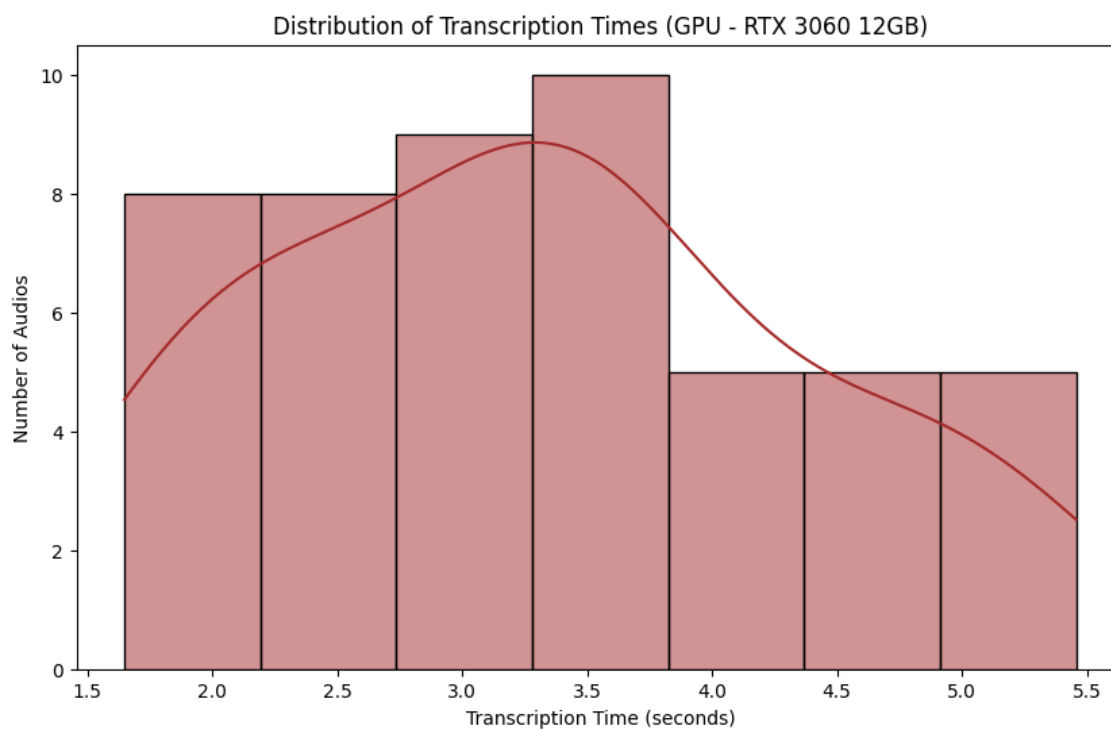


Figure 1.3: Transcription Times (GPU)

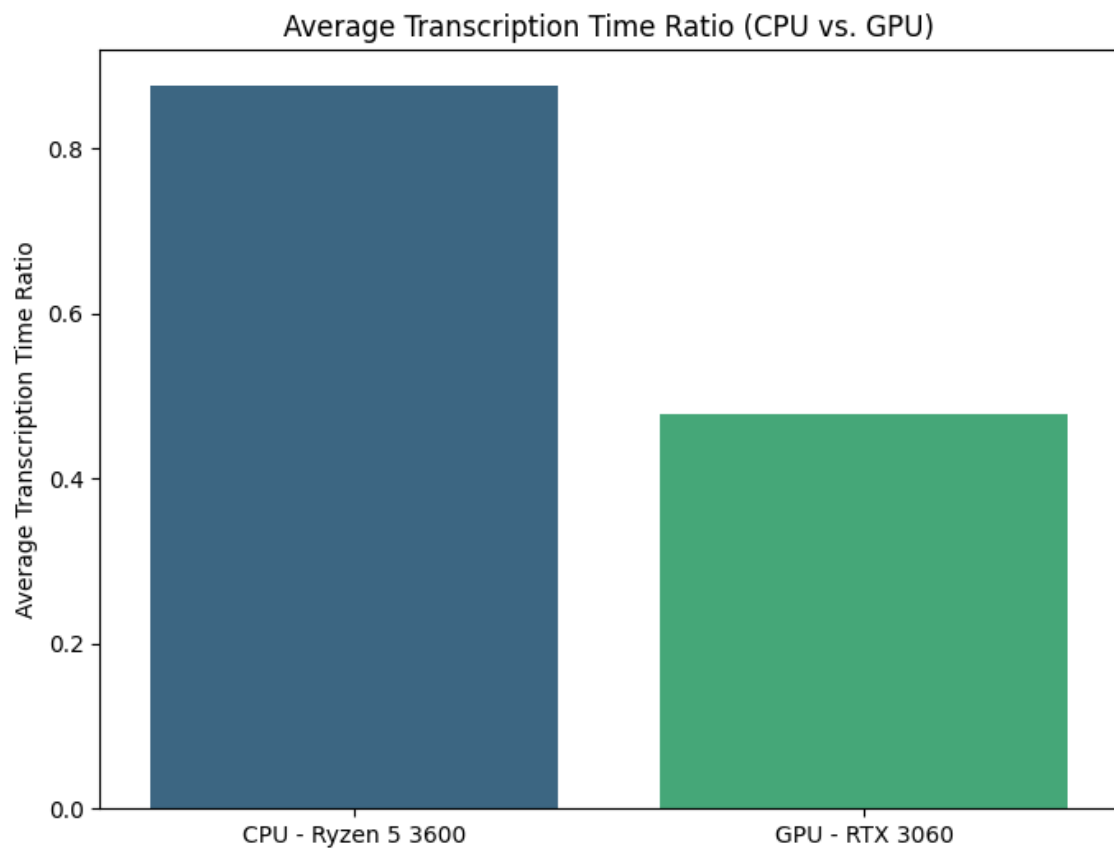


Figure 1.4: Average Transcription Time Ratio comparison

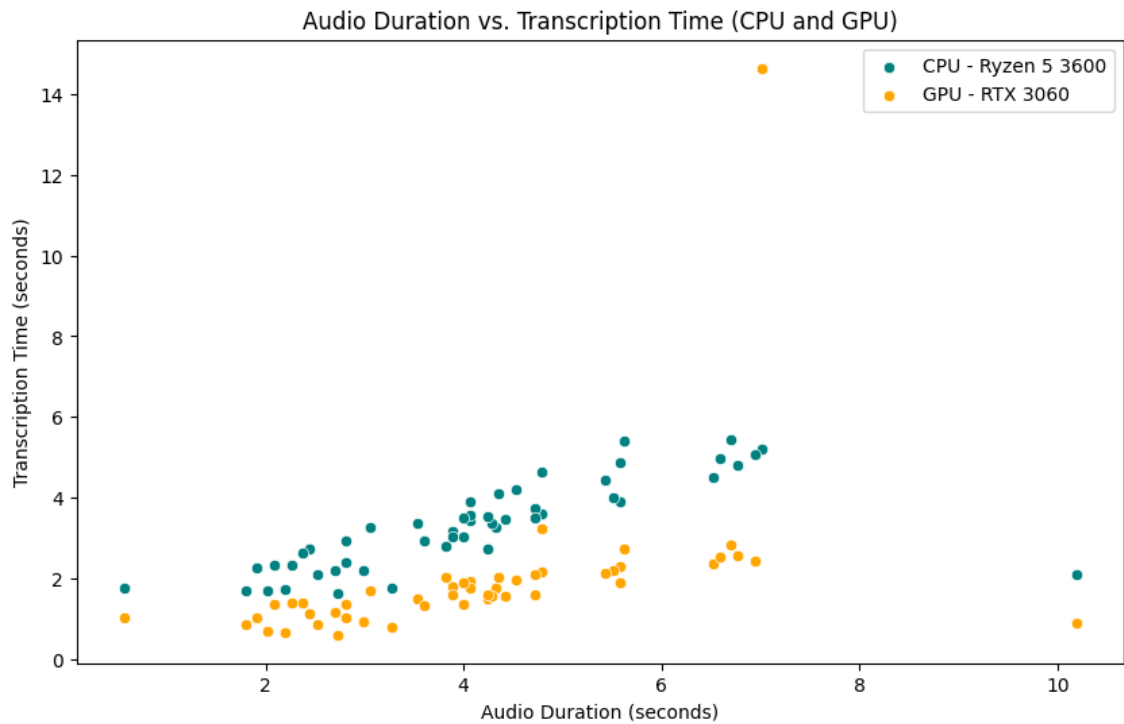


Figure 1.5: Audio duration vs Transcription Times compared

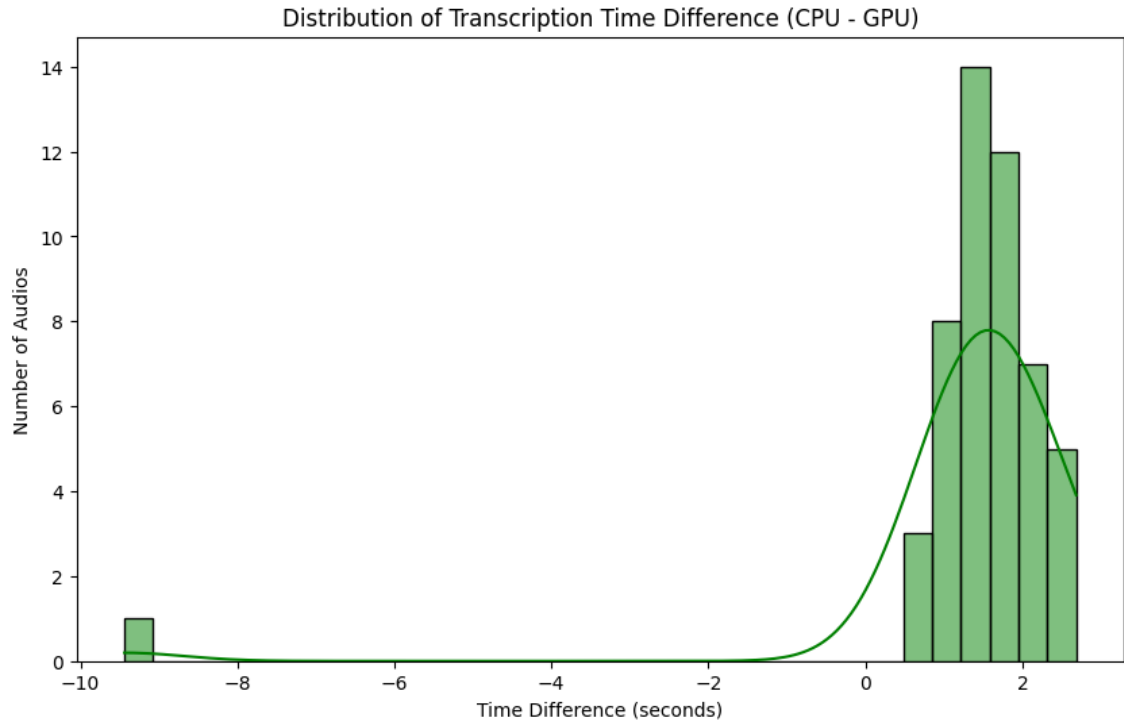


Figure 1.6: Transcription Time difference CPU GPU

Chapter 2

Literature Review

After the need for a robust ASR increased in the industry, [1]Radford et al proposed increasing the robustness of ASR models through weakly supervised learning. This meant that the data used was not up to golden standard and included noise. Thus, the model was able to transcribe speech even in noisy scenarios. This would later end up becoming the widely known Whisper by Open-AI. Many authors then furthered this research by building on the concept of training a model on large weakly supervised datasets or simply building up on Whisper. For example [2] Zhao et al uses the concept of zero-shot learning, [3] Peng et al tries to mimic and surpass Whisper's performance through e-branchformer and [4] Georgila et al used data augmentation to fine tune Whisper for low resource languages. During our research review we came across multiple brilliant approaches to achieving smaller WER for Speech to text systems, specifically for lower language resources. However, there was a lack of focus on Urdu and a resource constrained environment. The following Research items outlines the research we reviewed to identify the research gap and gain a better understanding of applicable models and possible solutions to our use-case.

2.1 Related Research

We did research on more than 15 research papers but we chose the 7 which are the most relevant to our studies. Following are the 7 research papers we studied :

- Robust Speech Recognition via Large-Scale Weak Supervision Whisper AI [1]
- Automatic speech recognition (ASR) with Whisper: Testing Per-formances in Different Languages. [2]
- Code-Switched Urdu ASR for noisy Telephonic Environment Using Data Centric approach with Hybrid HMM and CNN- TDNN[5]

- Better and Faster Open Whisper-Style Speech Models based on E-Branchformer[3]
- A PRISMA-driven Review of Speech Recognition based on English, Mandarin Chinese, Hindi and Urdu Language. [6]
- Improvement in Automatic Speech Recognition of South Asian accent Using Transfer Learning of DeepSpeech2.[7]
- Learning Cross-lingual Mappings for Data Augmentation to Improve Low-Resource Speech Recognition. [8]

2.1.1 Robust Speech Recognition via Large-Scale Weak Supervision Whisper AI

The paper [1] presents Whisper, a speech recognition system built on the concept of large-scale weak supervision. Whisper is trained on 680,000 hours of multilingual and multi-task supervised audio data from the internet. It focuses on zero-shot transfer learning without fine-tuning. The approach improves on more conventional methods by eliminating the need for adjustments in the dataset. It achieves near-human accuracy across diverse environments. The paper compares Whisper's performance to both traditional supervised models and human transcription standards which highlights its robustness in out-of-distribution settings. The system's key points are: Multitask learning, multilingual capabilities, and scaling large datasets.

2.1.1.1 Summary of the research item

This research focused on zero-shot transfer learning in the context of Automatic Speech Recognition (ASR). Whisper models can generalize to new tasks and datasets without explicit fine-tuning. They achieve this through training on large amounts of diverse, multilingual audio data. Key findings include Whisper's competitive performance compared to supervised systems and its robustness across various languages and acoustic environments. The method employs large-scale weak supervision, using open-source data from the internet and creates a general-purpose ASR system that approaches human-level robustness and accuracy. Whisper also used techniques such as text normalization and zero-shot evaluation, focusing on generalization beyond conventional benchmark datasets.

2.1.1.2 Critical analysis of the research item

Strengths:

- **Scale:** Because of its unparalleled dataset size, Whisper is able to achieve stronger cross-language and cross-environment generalization.
- **Zero-shot Performance:** The zero-shot evaluation shows a lot of robustness, easily outperforming previous supervised models in out-of-distribution scenarios.
- **Text Normalization:** The Whisper models reduce non-semantic errors during transcribing thanks to extensive normalizing techniques, which reduces WER (Word Error Rate) on a variety of datasets.

Weaknesses:

- **Model Complexity:** The model may not be as accessible to smaller research groups or enterprises without access to high-performance computers due to its heavy reliance on large datasets and computational resources.
- **Lack of Fine-tuning:** Whisper may do worse than refined models on some high-resource tasks, including VoxPopuli. This suggests that Whisper's efficacy may be limited in some domains by the lack of appropriate training for particular datasets.

2.1.1.3 Relationship to the proposed research work

Whisper's zero-shot approach to speech recognition aligns with our goal of maintaining high accuracy, especially in multilingual environments. The generalization capabilities demonstrated by Whisper models can inform our work on optimizing Whisper AI for Urdu. Specifically, we can adapt the use of large-scale weak supervision and text normalization techniques for Urdu language processing tasks. Moreover, Whisper's handling of diverse datasets suggests that similar approaches can be applied to improve performance on underrepresented languages like Urdu.

2.1.2 Automatic speech recognition (ASR) with Whisper: Testing Performances in Different Languages.

This research [2] examines how well OpenAI's Whisper ASR model performs for both healthy participants and those who are depressed in three different languages: English, Italian, and Russian. The amount of correct, deleted, substituted, and inserted words as well as the Word Error Rates (WER) for participants reading a standard fable in each of these languages are among the important metrics used in the study to assess Whisper's transcription accuracy.

2.1.2.1 Summary of the research item

In this study, audio recordings from 226 participants—who were categorized by language and mental health status (healthy vs. depressed)—were transcribed using Whisper. The fable "The Northern Wind and the Sun" by Aesop was assigned to the participants to read in their native tongues. Standard criteria, including accurate words, deletion, replacement, insertion, and WER, were used to assess Whisper's transcribing performance. Overall, the results showed good transcription accuracy, with just minor differences between people who were sad and those who were in good condition. Russian had a greater insertion rate, while English and Italian displayed larger substitution errors. All languages had consistently low WER.

2.1.2.2 Critical analysis of the research item

Strengths:

- **Cross-linguistic application:** Whisper's performance in three different languages shows its robustness in multilingual ASR systems.
- **Comprehensive dataset:** The use of both healthy and depressed participants provides valuable insights into how mental health can influence speech transcription accuracy.
- **Accuracy metrics:** The detailed analysis of transcription errors (correct, deleted, substituted, and inserted words) alongside WER makes the evaluation thorough.

Weaknesses:

- **Clinical group bias:** The study found that Whisper performed better for healthy participants in English and Italian, but there was less difference in Russian. The potential bias in transcription for clinical groups requires further exploration.
- **Language variations:** While Whisper handles multiple languages, the higher substitution and insertion rates in certain languages, such as Russian, suggest room for improvement in its handling of less commonly spoken languages.

2.1.2.3 Relationship to the proposed research work

This study is quite relevant to our suggested work on resource-constrained Whisper for Urdu optimization. Whisper's multilingual capabilities can direct the creation of a strong ASR for Urdu. The techniques employed here for error analysis, such as tracking particular word-level errors and analyzing WER, can be modified to evaluate and enhance Whisper's performance on Urdu language tasks.

2.1.3 Code-Switched Urdu ASR for noisy Telephonic Environment Using Data Centric approach with Hybrid HMM and CNN-TDNN

This research [5] discusses the performance of a proposed Automatic Speech Recognition (ASR) system for code-switched Urdu in noisy telephonic environments. Hidden Markov Models (HMM) and Deep Neural Networks (DNN) are used to create the ASR system. Specifically, a Convolutional Neural Network-Time Delay Neural Network (CNN-TDNN) with a Lattice Free Maximum Mutual Information (LF-MMI) objective function is used. With limited resources, the system strives for a low Word Error Rate (WER) and Sentence Error Rate (SER) while managing the intricacies of code-switched Roman Urdu.

2.1.3.1 Summary of the research item

The proposed ASR system tries to address the challenges of building a robust code-switched ASR model for telephonic environments. The researchers created a data-centric approach, collecting code-switched Urdu telephonic data and training the ASR model on it. The aim is to develop a system that can handle code-switched Roman Urdu with high accuracy in a call center setting. But also utilize minimal resources in terms of time, computing power, and memory. The model satisfies the aim of a WER below 10% and SER below 30%, and it can achieve an accuracy of 95% with only 10 hours of labeled data.

2.1.3.2 Critical analysis of the research item

Strengths:

- **Data-Centric Approach:** The focus on data collection and quality labeling ensures the system is built on strong foundations, emphasizing the importance of data over model tweaking.
- **Resource Efficiency:** The system is designed to be deployed in environments with limited computational resources, making it practical for call center applications in developing regions.
- **Performance:** The use of CNN-TDNN and LF-MMI for acoustic modeling enables the system to handle noisy, telephonic audio while achieving strong WER and SER performance.

Weaknesses:

- **Scalability:** The reliance on a smaller dataset (10 hours of labeled data) could limit the system’s performance on larger-scale applications or more diverse telephonic data.
- **Code-Switching Complexity:** While the system is optimized for Roman Urdu, the varying linguistic and acoustic patterns in different Urdu dialects or other code-switched languages could introduce additional complexities not addressed in the study.

2.1.3.3 Relationship to the proposed research work

This research is directly relevant to our goal of optimizing Whisper for Urdu in resource-constrained environments. The hybrid approach of using CNN-TDNN with LF-MMI could serve as a blueprint for developing an ASR system that efficiently handles Urdu in noisy environments. Additionally, the focus on reducing WER and SER aligns with our objectives of improving transcription accuracy while keeping computational demands low. Lessons from this study on data collection, particularly the emphasis on quality over quantity, will inform our own data-centric approach to optimizing Whisper for Urdu.

2.1.4 Better and Faster Open Whisper-Style Speech Models based on E-Branchformer

This Research Item [3] discusses the development of OWSM v3.1, a series of Open Whisper-Style Speech Models. The authors aimed to reproduce and enhance the performance of OpenAI’s Whisper model using publicly available data and an open-source toolkit. The key innovation in OWSM v3.1 is the introduction of E-Branchformer encoders, which outperform previous Transformer-based models used in prior OWSM versions. This research highlights how OWSM v3.1 achieves state-of-the-art performance across various speech processing tasks while also improving inference speeds by up to 25%. The study emphasizes the scalability and efficiency of OWSM v3.1 in large-scale speech recognition and translation tasks.

2.1.4.1 Summary of the research item

The OWSM v3.1 model series is an extension of previous Whisper-based models, focused on improving performance using E-Branchformer. This captures local and global information more effectively than standard encoders. The researchers tested their models on benchmarks for Automatic Speech Recognition (ASR) and Speech Translation (ST) tasks and found significant improvements in both performance and efficiency. They report better results on English and multilingual ASR and Speech Translation (X-to-English and

English-to-X) tasks compared to previous models, with OWSM v3.1 showing a reduction in Word Error Rates (WER) across various datasets like LibriSpeech and TEDLIUM.

2.1.4.2 Critical analysis of the research item

Strengths:

- **Efficiency:** The model achieves up to **25% faster inference** compared to its predecessors, making it highly efficient for deployment.
- **Scalability:** The study demonstrates the scalability of **E-Branchformer** across different model sizes (small, base, and medium), providing flexibility depending on the task’s computational requirements.
- **Open-source:** By using open datasets and releasing pre-trained models, the study promotes transparency and reproducibility in speech model research.

Weaknesses:

- **Data limitations:** The model’s performance in low-resource languages, like **Persian** and **Estonian**, is still suboptimal, likely due to the limited availability of training data for these languages.
- **Training complexity:** Training models for long-form ASR is challenging, and the models can struggle with convergence, especially when dealing with the 30-second audio chunks used for Whisper-style training

2.1.4.3 Relationship to the proposed research work

This research aligns closely with our work on optimizing Whisper AI for resource-constrained environments. The improvements in E-Branchformer can inform our strategy for scaling Whisper for Urdu, especially in low-resource scenarios. Moreover, OWSM v3.1’s focus on multilingual ASR and faster speeds gives valuable techniques for increasing both efficiency and performance of our Urdu ASR. The WER analysis and the model’s ability to handle multiple languages also highlight methods for improving our model’s generalization across diverse Urdu dialects.

2.1.5 A PRISMA-driven Review of Speech Recognition based on English, Mandarin Chinese, Hindi and Urdu Language

The paper [6] conducts voice recognition systems for four languages—English, Mandarin Chinese, Hindi, and Urdu—are reviewed using the PRISMA method. This study

compares various languages' approaches to Automatic Speech Recognition (ASR), with a particular emphasis on Urdu, in terms of development, dataset usage, feature extraction techniques, and deep learning-based methodologies. For Urdu, the study review period runs from 2015 to 2022; for the other three languages, it runs from 2018 to 2022.

Using benchmarks and datasets like HKUST (Mandarin Chinese), Librispeech (English), and other smaller Urdu speak datasets, the work employs systematic review processes. It also brings attention to the disparities in the development of ASR systems for the Urdu language and those with better resources.

2.1.5.1 Summary of the research item

After 176 papers are reviewed, 47 are selected for further consideration based on their applicability to the ASR pipeline. The dataset analysis reveals major differences in speech duration, speaker count, and vocabulary size. Urdu had a lot smaller dataset size than Mandarin and English. Due to the scarcity of training data, Urdu has a poorer accuracy for spontaneous speech recognition.

The primary conclusion is that, as a result of the limited data set and the application of traditional models like HMMs and LDA, the Word Error Rate (WER) for Urdu remains greater than that of other languages. Deep Neural Networks (DNNs) on the other hand, such Conformer and LSTMs, perform better for Mandarin and English, highlighting the technological divide between the two languages.

2.1.5.2 Critical analysis of the research item

Strengths:

- **Comprehensive comparative framework:** The PRISMA-driven method provides a systematic approach to comparing ASR systems across different languages.
- **Focus on Urdu:** The review brings much-needed attention to the resource limitations in Urdu ASR systems and the need for larger datasets.
- **Insights into WER differences:** By analyzing WER, the study highlights specific challenges for Urdu ASR, which suffers from high error rates due to the scarcity of data.

Weaknesses:

- **Limited coverage of deep learning techniques for Urdu:** While the paper highlights improvements in DNN-based systems for English and Mandarin, it shows

that **Urdu ASR systems** have not yet fully adopted advanced deep learning models, which limits the potential for improving WER.

- **Data disparity:** The lack of large datasets for Urdu remains a significant limitation, which the paper acknowledges but does not offer concrete solutions for overcoming this challenge.

2.1.5.3 Relationship to the proposed research work

This review is highly relevant to our goal, which is to optimize Whisper for Urdu in contexts with limited resources. The results define important WER and dataset size standards that inform our strategy for enhancing Urdu ASR performance. By using cutting-edge models like Conformer or Transformers to close the performance gap, we can contribute to the advancement of Urdu ASR systems. Furthermore, the Urdu ASR's reliance on feature extraction techniques like MFCC and Log Mel Spectrograms points to areas where Whisper might be improved.

2.1.6 Improvement in Automatic Speech Recognition of South Asian Accent Using Transfer Learning of DeepSpeech2

The research item [7] discusses the improvements in automatic speech recognition (ASR) for non-native English accents, specifically focusing on South Asian accents. The study builds upon the existing DeepSpeech2 model and explores how transfer learning can enhance the accuracy of ASR systems for non-native accents. The research aims to address the challenges faced by ASR systems, which are often trained on datasets featuring native English accents, leading to reduced accuracy for non-native speakers.

2.1.6.1 Summary of the research item

By adjusting the DeepSpeech2 model, the study offers a way to enhance ASR performance for South Asian dialects. Using a portion of the Common Voice dataset that exclusively contains South Asian accents, the researchers retrained the model. In order to maximize feature learning for these accents, three alternative layer topologies are evaluated. As evaluation criteria, the study uses Word Error Rate (WER), Match Error Rate (MER), and Word Information Lost (WIL). WER for non-native accents was significantly reduced in the final model, falling from 43% to 18.08%.

2.1.6.2 Critical analysis of the research item

Strengths:

- **Transfer learning approach:** The use of transfer learning to adapt a pre-trained model for non-native accents is innovative and highly effective in addressing the performance gap.
- **Layer configuration experiments:** Testing different model configurations allows for a better understanding of how deep learning models can be optimized for specific accent groups.
- **Improved WER:** The reduction in WER to 18.08% is a strong indicator of the model's ability to generalize well to South Asian accents.

Weaknesses:

- **Limited dataset:** The study used a subset of the **Common Voice dataset**, which may not fully represent the diversity of South Asian accents, leading to potential limitations in generalizing the model's findings to other non-native accents.
- **Focus on English ASR:** While the study effectively addresses non-native English accents, it does not explore how the findings could be applied to other languages spoken by South Asian speakers.

2.1.6.3 Relationship to the proposed research work

Our objective of refining Whisper AI for Urdu and other non-native speech recognition jobs is in line with this research. Whisper's performance for Urdu ASR can be bettered by applying transfer learning to change pre-trained models, specifically for speakers with varying regional accents. Furthermore, the emphasis on layer construction and fine-tuning is helpful for creating an ASR system for Urdu that is precise and uses fewer resources, using comparable transfer learning techniques to lower WER and better system performance as a whole.

2.1.7 Learning Cross-lingual Mappings for Data Augmentation to Improve Low-Resource Speech Recognition

This research [8] studies an end-to-end cross-lingual voice recognition system that uses data augmentation and mapping models to handle low-resource languages is covered in

this study. The system is primarily concerned with translating voice data between languages. It is especially helpful for low-resource languages such as Tamil, Telugu, Cebuano, and Javanese. The essential contribution comes in generating ciphered transcriptions using source language ASR systems and applying mapping models to complement the destination language ASR systems.

2.1.7.1 Summary of the research item

For the purpose of mapping posterior distributions from source language ASR models to target language ASR models, the paper presents a multi-encoder, single-decoder (MESD) architecture. By translating source language posteriors into the target language, cross-lingual speech recognition is made possible without the need for parallel data. Thirty hours were chosen for training from each language in the BABEL dataset. The method is focused on data augmentation. The audio from the source language is transliterated into the target language to produce ciphered text used to retrain the target ASR system. Character error rate (CER) improvements of up to 28.5% as compared to monolingual systems are shown by the results.

2.1.7.2 Critical analysis of the research item

Strengths:

- **Cross-lingual capability:** The ability to map between languages without parallel data offers a significant breakthrough for low-resource speech recognition tasks.
- **Data augmentation:** The integration of ciphered text as an augmentation technique provides a novel way to increase training data for under-resourced languages, demonstrating notable improvements in CER.

Weaknesses:

- **Mapping accuracy:** The accuracy of the mapping models varies between language pairs, with some languages (e.g., Tamil and Telugu) performing worse than others (e.g., Cebuano and Javanese), likely due to the differences in character set and linguistic structure.
- **Transliteration limitations:** The system's performance deteriorates for longer utterances, suggesting that the method may struggle with more complex speech patterns in low-resource languages.

2.1.7.3 Relationship to the proposed research work

Our objective of optimizing Whisper for Urdu ASR in low-resource environments is greatly impacted by this research. The application of data augmentation techniques along with the mapping model can enhance Whisper’s functionality for Urdu. We could improve Whisper’s capacity to generalize across a range of Urdu dialects and accents by utilizing comparable transliteration techniques. Furthermore, the multi-encoder, single-decoder (MESD) design offers a viable approach to addressing cross-lingual difficulties and might be modified to enhance Whisper’s functionality for Urdu speakers with diverse linguistic backgrounds.

2.2 Analysis Summary of research items

The following table outlines the research we reviewed to identify the research gap and gain a better understanding of applicable models and possible solutions to our use-case. Each entry summarizes Proposed Problem, Strengths, and limitations of the existing studies, providing a comprehensive overview of the current landscape. By analyzing these studies, we aim to highlight areas where further exploration is needed and establish a foundation for our proposed solutions.

Ref	Proposed Problem/Solution	Methodology	Strengths	Weaknesses
[1]	Robust Speech Recognition using Large-Scale Weak Supervision. Whisper AI	Trained on huge data using weak supervision for zero-shot ASR. Tested across diverse datasets for robust generalization.	Large dataset improves language generalization Strong zero-shot performance Effective text normalization reduces errors.	High complexity limits accessibility. Lacks fine-tuning for specific tasks.

[2]	Automatic speech recognition (ASR) with Whisper and testing its performances in Different Languages.	Developed with E-Branchformer encoders for ASR and translation. Tested on LibriSpeech and TEDLIUM to reduce WER and improve efficiency.	Strong cross-linguistic performance Diverse dataset includes both healthy and depressed participants Thorough accuracy analysis with WER and error types.	Potential bias in transcription accuracy for clinical groups Higher error rates in less common languages like Russian.
[5]	A robust ASR system for Urdu in call centers addressing challenges of low-resourced languages and code-switching with English.	Hybrid CNN-TDNN with LF-MMI trained on telephonic data. Optimized for low WER and SER using limited labeled data.	Data-centric approach ensures quality Efficient for deployment in resource-limited environments Strong performance with CNN-TDNN and LF-MMI in noisy audio.	Small dataset may limit scalability Challenges with code-switching and dialect variations.
[3]	Using E-Branchformer to improve the performance and efficiency of open-source speech models like Whisper without adding extra data.	Fine-tuned DeepSpeech2 using transfer learning with Common Voice data, tested with WER, MER, and WIL metrics.	25% faster inference for efficient deployment Scalable across different model sizes Open-source promotes transparency and reproducibility	Suboptimal performance in low-resource languages Training complexity in long-form ASR with 30-second audio chunks.

[6]	Investigates the development of Large Vocabulary Continuous Speech Recognition (LVCSR) systems, comparing Urdu with English, Mandarin Chinese, and Hindi using various acoustic and language models.	Used a multi-encoder, single-decoder architecture for posterior mapping, with ciphered transcriptions for data augmentation. Trained on BABEL data.	Systematic PRISMA-driven comparison of ASR systems Highlights resource limitations in Urdu ASR Provides insights into WER challenges for Urdu.	Limited adoption of deep learning for Urdu ASR Acknowledges but doesn't solve the data scarcity issue.
[7]	Fine-tuning the DeepSpeech2 model for South Asian accents using a novel loss function and dual pipeline architecture.	Combined Whisper with voice conversion (VQMIVC, DiffCV) to reduce WER for non-native accents by converting voices to native-like speech.	Effective transfer learning for non-native accents Layer configuration experiments optimize performance Improved WER (18.08%) for South Asian accents.	Limited dataset may not capture full accent diversity Focuses only on English ASR, not other languages.
[8]	The paper investigates whether Open-AI's Whisper ASR models can perform speech-based in-context learning to reduce word error rates in Chinese dialects without retraining.	Systematic review comparing ASR systems in English, Mandarin, Hindi, and Urdu, focusing on dataset sizes, features, and model differences.	Cross-lingual capability enhances low-resource speech recognition Ciphered text data augmentation improves training data and reduces CER.	Mapping accuracy varies by language pair, with some pairs underperforming Performance declines for longer utterances, indicating challenges with complex speech.

Table 2.1: Summary of Research

Chapter 3

Proposed Approach

To optimize the transcription process using Whisper AI, we propose a system that leverages both CPU and GPU-based processing environments to handle large-scale Urdu datasets. In our tests, we observed significant differences in processing times: transcription of a 5-second audio file took 25 seconds on a CPU and 5 seconds on a GPU. The proposed system will focus on utilizing GPU resources to maximize efficiency while maintaining accuracy.

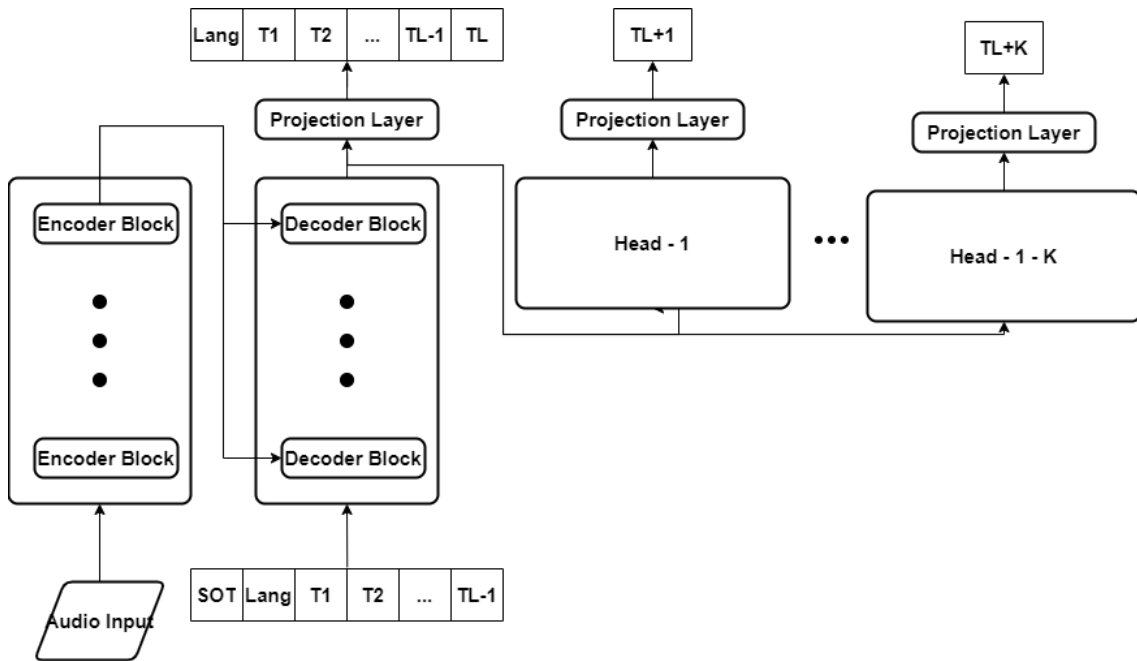


Figure 3.1: Multi-Head Attention

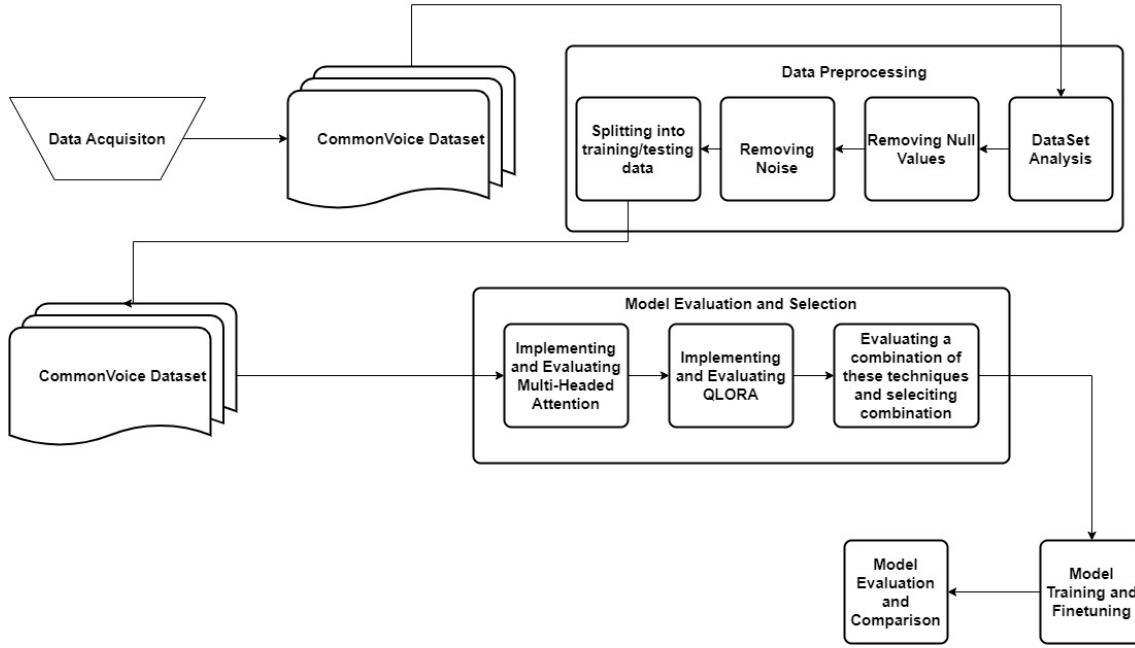


Figure 3.2: Workflow

3.1 System Configuration

3.1.1 GPU Acceleration

Since GPU processing yields a 5x improvement over CPU, the system will prioritize GPU-based transcription for high-demand environments, particularly in call centers where thousands of hours of audio need to be processed daily. The system will be built to ensure GPU resources are utilized to their full extent for real-time transcription. However, it must also be kept in mind that such systems might also be readily available in such scenarios and design a system that minimises the spec requirements of GPU.

3.1.2 Hybrid Approach (CPU + GPU)

For smaller workloads or when GPU resources are limited, the system will fall back on CPU processing to ensure continuity. This hybrid approach guarantees operational flexibility.

3.2 Optimized Whisper Model

3.2.1 Whisper-Mini

A lightweight version of Whisper will be employed, fine-tuned for Urdu with the aim to strike a balance between speed and accuracy. The model will be optimized to reduce Word Error Rate (WER) for Urdu-specific phonetic structures.

3.2.2 Resource-efficient Modifications

Techniques like multi-head attention will be explored to further reduce inference times without sacrificing accuracy

3.2.3 QLoRA (Quantized Low-Rank Adaptation)

: To improve the efficiency of model training and inference, we will implement QLoRA, which allows for efficient fine-tuning of large language models with significantly reduced memory requirements. QLoRA first quantizes the model, reducing the numerical precision of model parameters. This minimizes the memory needed to run these models. For example, instead of using 16 bits to store a parameter, we can represent it using 8 bits, which is essential for real-time performance.

3.2.3.1 Combining Techniques

With QLoRA, we can fine-tune the model without sacrificing precision by training smaller low-rank matrices that approximate the original parameters. This process speeds up the fine-tuning and also makes it feasible for those with limited computational resources. For example, while a full fine-tuning of a 70 billion parameter model typically requires 780GB of GPU memory, QLoRA allows us to achieve this with only 48GB, allowing easier access to advanced model fine-tuning.

3.3 Workload Balancing

3.3.1 Parallel Processing

The system will use parallel processing techniques to distribute the workload across multiple GPUs or CPUs, depending on resource availability.

3.3.2 Batch Processing

Audio files will be transcribed in batches, ensuring that multiple files are processed simultaneously, further reducing the overall time required for large datasets.

3.4 Methods and Workflow Explanation

3.4.1 Multi-Head Attention (Figure 3.1)

The **Multi-Head Attention** mechanism, inspired by transformer models, will be employed in Whisper-Mini to improve accuracy and reduce processing time. This technique allows the model to focus on different parts of the input audio at once, resulting in better representation of Urdu phonetic features and improved transcription performance. Multi-Head Attention ensures that the model captures both short-term and long-term dependencies in speech.

- **Attention Heads:** Each attention head focuses on different parts of the audio input, making it easier to capture subtle variations in pronunciation and tone specific to Urdu.
- **Key, Query, and Value Vectors:** The mechanism generates these vectors to compute how much focus each word or sound in the audio should receive, resulting in more accurate transcriptions.

3.4.2 Workflow (Figure 3.2)

The workflow diagram outlines the transcription process, starting from audio input to the final transcribed output:

1. **Audio Input:** The system receives Urdu audio data, either in real-time or from a pre-recorded source.
2. **Preprocessing:** The audio data is processed to remove noise and irrelevant audio segments. This step ensures that only the essential data is fed into the model.
3. **Model Inference:** The Whisper-Mini model is applied to the preprocessed audio. For large datasets, the system dynamically allocates GPU resources for faster inference. If GPU resources are limited, the system can fallback to CPU processing.

4. **Transcription:** The output of the model is the transcribed text, which undergoes post-processing to correct any linguistic nuances, improve punctuation, and ensure text normalization (to handle Urdu’s script properly).
5. **Output Delivery:** The transcription is either sent directly to the user via a web interface or stored in a database for further use.

This workflow ensures smooth integration between different components, allowing for efficient transcription in both real-time and batch modes.

3.5 Performance Evaluation

- **Transcription Speed:** The goal is to reduce transcription time on GPUs to near real-time performance (1:1 or better), ensuring 5 seconds of audio are transcribed in 5 seconds or less.
- **Accuracy:** The system will be evaluated on transcription accuracy for Urdu using **WER (Word error rate) and CER (Character Error Rate) metrics**. Fine-tuning will be focused on reducing errors specific to Urdu’s phonetic and linguistic characteristics.

This approach ensures both scalability and efficiency, making it feasible for real-time applications in resource-constrained environments. The diagrams provide a clear visual representation of how multi-head attention and the overall workflow improve both performance and accuracy.

Chapter 4

Conclusions

We have provided a thorough strategy for optimizing Whisper AI for effective transcription of Urdu audio in resource-constrained settings in our suggested technique. We may achieve a large reduction in transcribing time without sacrificing accuracy by utilizing advanced approaches like multi-head attention, exploiting GPU acceleration, and putting in place a hybrid CPU-GPU framework. Because Urdu has particular linguistic and phonetic features, the Whisper-Mini model, which has been adjusted for the language, ensures a decreased Word Error Rate (WER) and Character Error Rate (CER).

Scalability is ensured by the integration of batch and parallel processing, allowing the system to handle massive datasets in real-time, like those produced in call center situations. Furthermore, the multi-head attention and workflow diagrams help to clarify how the model operates and maximizes performance. Scalability is ensured by the integration of batch and parallel processing, allowing the system to handle massive datasets in real-time, like those produced in call center situations. Furthermore, the multi-head attention and workflow diagrams help to clarify how the model operates and maximizes performance.

This solution offers a versatile, scalable, and resource-efficient system appropriate for a range of large-scale applications, in addition to improving the speed and accuracy of Urdu transcription. The method makes a significant contribution to the field of Automatic Speech Recognition (ASR) for low-resource languages like Urdu because it strikes a compromise between computing economy and linguistic correctness.

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